Crop Yield Prediction using Machine Learning

By

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Declaration

We do hereby declare that the project works presented in this report entitled, "**Crop Yield Prediction using Machine Learning**" are the results of our works. We further declare that the report has been compiled and written by us, and no part of this report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this report.

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Approval

I do hereby declare that the project works presented in this report, entitled "**Crop Yield Prediction using Machine Learning**", are the outcome of the original works carried out by **Md. Mahabub Jamil**, **Md. Mostafizur Rehman Emon**, **Md. Mahedi Hasan**, **Md Bipul Sarker** under my supervision. I further declare that no part of this report has been submitted elsewhere for the requirements of any degree, award, diploma, or other purposes except for publications. I further certify that the dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

This research is for our wonderful parents, whose constant support and encouragement have been the foundation of our academic career.

Acknowledgment

We would like to express our heartfelt gratitude to the almighty Allah who offered our family and us kind care throughout this journey. We are deeply thankful to the Bangladesh University of Business and Technology (BUBT) for providing us with such a wonderful environment to pursue our project. We would like to express our sincere gratitude to **Md Masudul Islam**, Assistant Professor, Department of CSE, BUBT. We have completed our project with his help. We found the project area, topic, and problem with his suggestions. He guided us with our study and supplied us with many project papers and academic resources in this area. He is patient and responsible. When we had questions and needed his help, he would always find time to meet and discuss with us, no matter how busy he was. We also want to give thanks to **Prof. Dr. Md. Ahsan Habib**, Professor & Chairman, Department of CSE, and all our teachers for providing a solid background for our studies and research thereafter. They have been a great source of inspiration to us, and we thank them from the bottom of our hearts. We would also like to acknowledge our team members for supporting each other and be grateful to our university for providing this opportunity for us.

Abstract

Choosing the right crop is fundamental to successful farming, directly impacting the consistency and volume of yields. Today's farmers, however, face a complex web of challenges from unpredictable weather and degrading soil quality to economic volatility, making this critical decision increasingly difficult. An unsuitable crop choice can lead to diminished harvests, financial strain, and inefficient use of valuable resources. To help farmers navigate these complexities, this research introduces a predictive model built on powerful machine learning algorithms: XGBoost, Random Forest, and Decision Trees. This model analyses key environmental variables, including soil nutrient levels (nitrogen, phosphorus, potassium), temperature, humidity, pH, precipitation, and soil type, to recommend the most suitable crops for a specific piece of land. After processing a comprehensive dataset, our models effectively identified optimal crop types. We have translated this predictive power into a practical tool: an online interface that allows farmers to enter their local environmental data and receive immediate, data-driven crop recommendations. Among the algorithms tested, XGBoost demonstrated outstanding performance and accuracy. The ultimate goal of this project is to provide farmers with an intuitive, accessible system that empowers them to enhance their productivity, make the most of their resources, and adopt more environmentally sustainable practices.

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Chapter 1

1. Introduction

While agriculture is critical to feeding the world's people, it also faces many problems, from changing weather patterns and soil degradation to climate change. Such issues and the increasing demand for food make it even more critical for farmers to choose the best crops that can give their maximum output of food. In this work, we delve into modelling the best crop that can be grown in each field while utilizing different environmental variables like nutrient levels in the soil, temperature, humidity, precipitation, etc, using a machine learning-based approach [21]. The aim of this project is to assist farmers in using their data to make better operational decisions, enabling them to get the most out of their fields and waste as few resources as possible.

1.1. Problem Statement

Farming becoming less stable due to climate change means more farmers are growing produce in variable conditions, but that may come with some benefits. If not armed with field-level data on what crops should perform best in different circumstances, farmers stand to lose entire seasons of tens and hundreds of acres of crops. This squanders resources, and results in lower fulfilment production. This research work builds upon developing predictive system to suggest which crops can be grown based on different environmental variable inputs. And that would help farmers to grow more crops and better use of other resources.

1.2. Problem Background

People would know of the crops that they grew and harvested before, so now they would go out pick what was theirs. This, however, is a plan that does not take into consideration the continuing differences in weather conditions, humidity and soil quality. More extreme weather is the result of climate change, which farmers everywhere are left to cope with. This means they want tech that helps them more than ever make real-time decisions. This paper suggests the employment of machine-learning, to reduce and enhance crop selection operation, in order to provide farmers with reliable decision making based on data

1.3. Research Objectives

This research focuses on the critical challenge of accurately predicting crop yield. By leveraging historical data on soil properties and weather patterns, the project aims to build a robust predictive system. The core of this research involves applying and evaluating a suite of machine learning algorithms to analyse these complex factors and identify the most accurate model. The ultimate goal is to develop a practical tool that can empower farmers and agricultural stakeholders with data-driven insights, helping them make better decisions about resource management and crop selection.

1. Understand which soil and weather factors most affect how well crops grow.
2. Build a smart Machine Learning tool that suggests the best crops to plant based on local conditions.
3. Create a simple web app to give farmers real-time advice on what to plant.

1.4. Motivations

This study was inspired by the need for better and knowledge informed interventions in agriculture. With current data, farmers can predict the harvest results accordingly with machine learning. This will be helpful for farmers to optimize their yield and to reduce wastage. The main goal here is to make sure the tool we provide farmers with true and useful information.

1.5. Significance of the Research

This research is significant because it provides farmers with a smart, data-driven tool to help them choose the best crops for their specific land. By analysing soil and weather conditions, the system gives farmers reliable advice, reducing the guesswork and risk that often lead to poor harvests and financial loss. This helps farmers increase their crop yields, leading to more stable incomes and greater financial security. The tool also promotes more efficient farming. By recommending crops that are naturally suited to the local environment, it helps farmers save money on expensive resources like water and chemical fertilizers. This is also better for the environment, as it prevents the overuse of fertilizers that can pollute waterways and damage soil health for the future. By making farming more productive and sustainable, this research contributes to a more stable and secure food supply for a growing global population. Overall, it provides a simple, effective, and convenient solution that empowers farmers to make smarter decisions, protect the environment, and build a more resilient agricultural future.

1.6. Research Contribution

1. We used advanced computer models like XGBoost and Random Forest to give highly accurate and reliable crop suggestions.
2. The system gives advice based on specific soil and weather data, helping farmers grow more food while using less water and fertilizer.
3. We built a simple web app that gives farmers instant crop recommendations, so they don't need to be tech experts to use it.

1.7. Complex Engineering Analysis

1.7.1. Complex Engineering Problems

P1: Depth of knowledge required: This project requires a deep understanding of several different fields at once. It's not just about computer science; it also involves knowing about machine learning models, data analysis, and the basics of agriculture, like how soil nutrients and climate affect crop growth.

P2: Conflicting technical requirements: The project involves balancing goals that often conflict with each other. For instance, the prediction model needs to be highly accurate to be useful, but it also has to be simple and fast enough to work smoothly on a basic web app for farmers who need quick answers.

P3: No obvious solution: There isn't one single, clear-cut way to solve this problem. Recommending the right crop involves a deep analysis of complex data and requires trying out different computer models, like XGBoost and Random Forest, to see which one gives the most trustworthy advice.

P4: Unfamiliar issues: This project tackles issues that are not usually part of a computer science education. It forces an engineering approach to be applied to unfamiliar topics like agricultural science and environmental conditions, which is a new and challenging area for most developers.

P7: Interdependence: The project is made up of many smaller, connected sub-problems that all have to work together perfectly. This includes everything from collecting and cleaning the farm data to training the prediction models and building the final web application; a failure in one part can affect the whole system.

1.7.2. Complex Engineering Activities

A1: Range of resources: This project requires using a wide variety of resources to be successful. This includes everything from the agricultural datasets and machine learning software to the web development tools needed to create the final, user-friendly application for farmers.

A3: Innovation: The project is innovative because it applies modern computer science ideas to a very traditional field. Using complex machine learning to solve a real-world problem for farmers is a creative and new way to use engineering principles to make a difference.

A4: Consequences for society and the environment: The work done in this project has important consequences for both society and the environment. By helping farmers make better decisions, it can improve food security, while also promoting farming practices that are more sustainable and less harmful to the planet.

A5: Familiarity: This activity involves working in an area that is new and unfamiliar to most computer engineers. It requires learning about and applying technical skills to a completely different field—agricultural science—which is a significant challenge beyond typical software development.

1.8. Thesis/Project Organization

The remainder of this thesis is structured as follows: Chapter 1 provides an introduction to the project, including its goals, motivations, and contributions. Following this, Chapter 2 presents a review of similar research and an analysis of the problems this project addresses. Chapter 3 explains the system's design and the methods used to build it. Subsequently, Chapter 4 details how the system was built and how its different parts were connected. Chapter 5 offers a look at the test results and an evaluation of how well the models performed. Finally, Chapter 6 concludes the thesis with a final summary of the project, its limitations, and ideas for future work.

Chapter 2

2. Background Study

In this section, we present a detailed Literature Review on data-driven approaches in crop yield prediction. The following table summarizes the key studies that form the basis of this review, highlighting the datasets, methodologies, and principal findings that define the current state of the art.

2.1. Literature Review

In this part, we are summarizing some previous studies. In **Table 1** the summary of various crop yield prediction studies is presented.

Table 2.1. Summary of various Crop Yield Prediction studies

Study	Dataset/Database	Methods/Models	Result	Remarks
Feng et al. (2021) [1]	Winter wheat yield (2008–2019) in the US; MODIS NDVI, EVI; meteorological data.	Geographically and Temporally Weighted Neural Network (GTWNN), compared with GTWR, ANN, SVR.	GTWNN achieved the highest accuracy (, RMSE = 16.403), outperforming other models by capturing spatiotemporal non-stationarity.	First application of a GTWNN model to explicitly address the changing relationship s between predictors and yield across space and time.
Agarwal et al. (2021) [2]	Crop dataset with features like temperature, humidity, rainfall, soil type.	Hybrid approach using both Machine Learning (ML) and Deep Learning (DL) algorithms (DT, ANN, etc.).	The paper proposes a framework for predicting best crop production but does not present specific quantitative results for a single best model.	Provides a general overview and framework for applying AI algorithms to assist farmers, focusing on a hybrid ML/DL approach.
Wang et al. (2022) [3]	Winter wheat yield (2003–2017) in Henan, China;	Long Short-Term Memory (LSTM),	LSTM model achieved the best	Showcased the power of LSTM for time-series

	MODIS LAI products.	compared with RF, SVR, ANN.	performance (, RMSE = 522.3 kg/ha), demonstrating superiority over conventional ML methods using a single data factor (LAI).	analysis in yield prediction, even with a single input variable, highlighting its potential for data-scarce regions.
Hara et al. (2021) [4]	Review paper covering various crops and regions.	Review of Artificial Neural Network (ANN) models.	Discusses a wide range of independent variables (meteorological, soil, remote sensing VIs) used in ANN models for crop yield prediction.	Provides a comprehensive review of input variable selection for ANN-based crop yield forecasting, categorizing factors and discussing their limitations.
Shahhosseini et al. (2021) [5]	Corn yield (2010–2018) in the US Corn Belt; weather, soil, and management data.	Hybrid approach coupling the APSIM crop model with ML models (LASSO, LightGBM, RF, XGBoost).	The hybrid (APSIM+ML) approach reduced RMSE by 12–21% compared to ML models alone, demonstrating the value of integrating biophysical model outputs.	Systematically demonstrated that coupling a mechanistic crop model with ML algorithms significantly improves prediction accuracy by providing biophysically relevant features.
Bharadiya et al. (2023) [6]	Review paper (2011–2022) on various crops.	Review of ML, DL, and remote sensing techniques.	Identifies satellite technology (MODIS) and Vegetation Indices as the most frequently used remote sensing tools	A broad review comparing various techniques and highlighting the prominence of remote sensing data, particularly from satellite

			and attributes for crop yield forecasting.	platforms like MODIS.
Gupta et al. (2022) [7]	Review paper on various crops.	Review of ML frameworks, including feature selection methods like Particle Swarm Optimization (PSO) with SVM.	Highlights the improved performance of ML algorithms when combined with effective feature selection and extraction techniques.	Emphasizes the importance of feature engineering and selection (e.g., PSO-SVM) in building accurate and efficient ML-based prediction frameworks.
Batool et al. (2022) [8]	Tea crop yield (2016–2020) in Pakistan; weather, crop, soil, agro-management data.	Hybrid approach comparing the AquaCrop simulation model with ML models (Linear SVR, RF, DT, Bagging Regressor).	ML models, particularly Bagging Regressor (), significantly outperformed the calibrated AquaCrop model () for tea yield prediction.	Provides a case study of a hybrid simulation-ML approach for a perennial crop (tea), concluding that data-driven ML models can be more accurate than calibrated CSMs.
Khaki, Pham, & Wang (2021) [9]	Corn and soybean yield (2004–2018) across 13 US states; MODIS satellite data.	A novel deep transfer learning CNN called YieldNet for simultaneous multi-crop prediction.	YieldNet accurately predicted yields 1–4 months before harvest (MAE of 8.74% for corn, 8.70% for soybean) and outperformed single-crop models.	Pioneering work in multi-target regression for crop yield, demonstrating a computationally efficient framework for predicting multiple crops simultaneously.
Ji et al. (2021) [10]	Corn yield (2008–2018) in the US Corn Belt; MODIS-NDVI time series.	Phenological approach using multivariate regression models based on metrics	Phenological metrics (duration and rate of growth phases) were significantly	Demonstrated that physically meaningful phenological metrics derived from VI time

		extracted from NDVI curves.	correlated with yield and achieved high prediction accuracy (up to 0.72).	series can serve as powerful predictors for large-scale yield forecasting.
Khaki & Wang (2020) [11]	Corn and soybean yield (2016–2018) in the US Corn Belt; environmental data and management practices.	Hybrid CNN-RNN model, compared with LASSO, RF, and DFNN.	The hybrid CNN-RNN model significantly outperformed other models, achieving the highest correlation and lowest RMSE for both corn and soybean.	Developed a novel hybrid DL architecture where CNNs act as feature extractors for environmental data and RNNs model the temporal dependencies.
Shook et al. (2021) [12]	Soybean yield from the Uniform Soybean Tests Northern (USTN) dataset; genotype, weather, and management data.	Deep learning framework using stacked LSTMs with a temporal attention mechanism.	The LSTM-based models outperformed traditional models like LASSO and SVR, and the attention mechanism provided insights into critical weather periods.	A key study integrating genotypic data (molecular markers, pedigree) with weather variables in a deep learning model, using an attention mechanism for interpretability.
Yli-Heikkilä et al. (2022) [13]	Cereal crop yields (2018–2020) in Finland; Sentinel-2 time series data.	Temporal Convolutional Network (TCN), compared with Random Forest (RF).	TCN outperformed RF and was robust to the presence of cloudy pixels, suggesting it can learn cloud masking implicitly. Histogram-	Showcased TCN as a powerful and operationally lightweight method for large-scale prediction, and importantly, highlighted its robustness to data quality issues like cloud cover.

			based features performed best.	
Srivastava et al. (2022) [14]	Winter wheat yield (1999–2019) in Germany; weather, soil, and crop phenology data.	1D-CNN model, compared with DNN, XGBoost, and other ML baselines. SHAP was used for interpretability.	The proposed 1D-CNN model outperformed all baselines (7–14% lower RMSE). SHAP analysis identified key features and critical time windows.	A strong case for using 1D-CNNs to capture time dependencies in environmental data and a valuable application of SHAP for model interpretability in agriculture.
Gong et al. (2021) [15]	Tomato yield from three greenhouse datasets; environmental parameters and historical yields.	A novel DNN combining TCN and RNN.	The proposed TCN-RNN model achieved more accurate predictions (lower RMSE) than traditional ML methods and other standalone DL architectures.	Focused on the controlled environment of greenhouses and proposed a hybrid TCN-RNN architecture, demonstrating the power of combining different sequence modelling approaches.
Kuradusenge et al. (2023) [16]	Irish potato and maize yield in Musanze, Rwanda; historical weather and yield data.	Random Forest (RF), Polynomial Regression (PR), and Support Vector Regressor (SVR).	RF was the best-performing model for both crops, achieving values of 0.875 (potato) and 0.817 (maize).	A case study applying classical ML models in a data-scarce, low-to-middle income country, demonstrating the effectiveness of RF for local-scale prediction.
Sharifi (2020) [17]	Barley yield in Iran; Sentinel-2 (NDVI, EVI); TerraClimate (weather data).	Gaussian Process Regression (GPR), compared with BPNN, DT, KNN.	GPR performed best one month before harvest (, RMSE = 737 kg/ha).	Investigated the effect of different time intervals on yield prediction and found GPR to be the most effective model.

Bali & Singla (2021) [18]	43-year benchmark dataset for wheat in Punjab, India.	RNN with LSTM, compared with ANN, RF, and MLR.	RNN-LSTM model significantly outperformed others (RMSE of 147.12 vs. 540-915 for other models).	Demonstrated the efficacy of RNN-LSTM for time-series crop yield prediction, effectively handling the vanishing gradient problem.
Gupta & Nahar (2022) [19]	Crop dataset for an IoT-based system.	Two-tier ML approach: aKNCN for classification and ELM-mBOA for prediction.	The proposed aKNCN-ELM-mBOA model achieved the lowest error rates compared to other approaches.	Proposed a novel two-tier framework for both classification and prediction in a smart agriculture context.
Shaik et al. (2022) [20]	Rice production dataset for Kharif and Rabi seasons.	Random Forest (RF) and Linear Regression.	Provides a comparative analysis of the models for crop yield prediction.	A comparative study of ML algorithms on weather and crop datasets to identify impactful variables for yield.

2.2. Problem Analysis

Based on the literature review, we identified several prevailing problems and research gaps that persist in the field of data-driven crop yield prediction. First, many advanced deep learning models function as "black boxes," making it difficult to understand the reasoning behind their predictions. This lack of transparency is a major barrier to adoption, as stakeholders need to trust and interpret the factors driving a forecast. Techniques like SHAP analysis and attention mechanisms are being used to address this, but their application is not yet standard practice. Second, data quality and availability remain significant challenges. Issues such as cloud contamination in satellite imagery require extensive pre-processing, and data scarcity in many regions limits the performance of data-hungry deep learning models. While some newer architectures like TCNs show robustness to noisy data, and transfer learning offers a solution for data scarcity, these are still active areas of research. Third, model scalability and generalizability are critical hurdles. Many empirical models developed for specific locations fail to generalize accurately when applied across different

geographical scales or under new climatic conditions not seen in the training data. Specialized architectures like GTWNN and hybrid science-guided AI approaches are being developed to tackle this, but it remains a key challenge. Finally, the vast majority of research focuses on a single-crop paradigm, requiring separate, inefficient models for each crop of interest. This overlooks the potential benefits of multi-target frameworks, like YieldNet, that can predict yields for multiple crops simultaneously, representing a significant and promising frontier for future work.

Chapter 3

3. Research Methodology

Project Methodology is following a structured approach to develop the crop yield prediction system. Its process begins with data analysis to identify the key features from the augmented dataset. The Python backend script is used to train and compare several machine learning models. XGBoost is used as the final prediction model for its high accuracy, while a Python app creates the user-friendly interface. Scikit-learn [23] handles the model training and evaluation, and Joblib saves the trained model securely for future use. Finally, the system undergoes testing and verification to make certain accuracy, reliability, and overall performance.

3.1. Proposed Framework

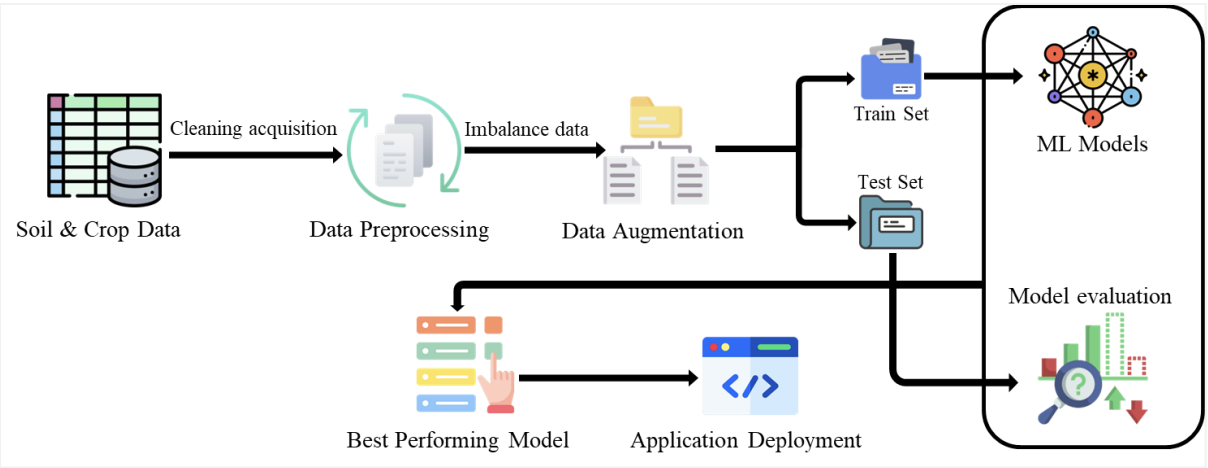

Figure 3.1. Crop Yield Prediction Proposed Architecture

Figure 3.1 illustrates the end-to-end workflow of the machine learning system, structured as a sequential pipeline designed to transform raw data into a deployable application. The process initiates with Data Acquisition, where raw datasets are ingested and immediately subjected to a Cleaning & Pre-processing phase to address inconsistencies. To tackle potential distribution issues, the workflow includes a specific step for handling Imbalanced Data followed by Data Augmentation to enhance the dataset's robustness. This refined data is then split into distinct Training and Testing Sets. The core modelling phase involves training various ML Models on the training data, followed by a rigorous Model Evaluation stage using the test set to assess performance metrics. Based on these evaluations, the system identifies the Best

Performing Model, which is selected for the final stage: Application Deployment, where the optimized model is integrated into a user-facing software interface.

3.2. Data Set Analysis

Table 3.1 shows at a glance of our dataset description.

Table 3.1. Summary of Crop Yield Prediction Datasets

Number of Original Data	2200
Number of Augmented Data:	11000
Number Of Features:	10 *(Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH_ Value, Rainfall, Moisture, Soil Type)*
Number of Result Features:	1(Crop)
Train Set:	80% (8800)
Test Set:	20% (2200)
Number of Crops:	22 *(Apple, Banana, Blackgram, ChickPea, Coconut, Coffee, Grapes, Jute, KidneyBeans, Lentil, Mango, MothBeans, MungBean, Muskmelon, Orange, Papaya, PigeonPeas, Pomegranate, Rice, Watermelon, Cotton, Maize)*

3.2.1. Data Pre-processing

The data pre-processing phase was critical to ensuring the quality and reliability of the inputs for the machine learning models. This process involved a sequential multi-step approach starting with data integration. Initially, the research utilized two distinct datasets containing agricultural and environmental records, comprising a total of 2200 raw samples. These two sources were consolidated and merged into a single, unified dataset to create a comprehensive base for analysis. To address the data requirements of deep learning and robust machine learning training, a Data Augmentation technique was subsequently applied. This step artificially expanded the dataset size significantly, resulting in a final count of 11000 records. This introduced necessary variability to help the models generalize better and prevent overfitting. Following augmentation, a rigorous Data Cleaning process was executed where the dataset was scanned for null or missing values, and incomplete records were removed. The final pre-processed dataset consists of 11 key attributes: the input features Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH_Value, Rainfall, Moisture, Soil Type, and Crop, along with the target variable Yield. This clean,

augmented dataset served as the foundation for all subsequent model training and evaluation.

3.3. Algorithm/ Model Analysis

The workflow illustrated in **Figure 3.2** begins with the prepared dataset, which is split into two parts: 80% for training and 20% for testing. The training data is used to build the XGBoost model, which learns patterns and relationships from the input features. The test data is then used to evaluate the model's performance through metrics such as accuracy, precision, recall, and other rating indicators. Finally, the evaluation results determine the model's effectiveness and reliability, producing the final outcome or prediction results.

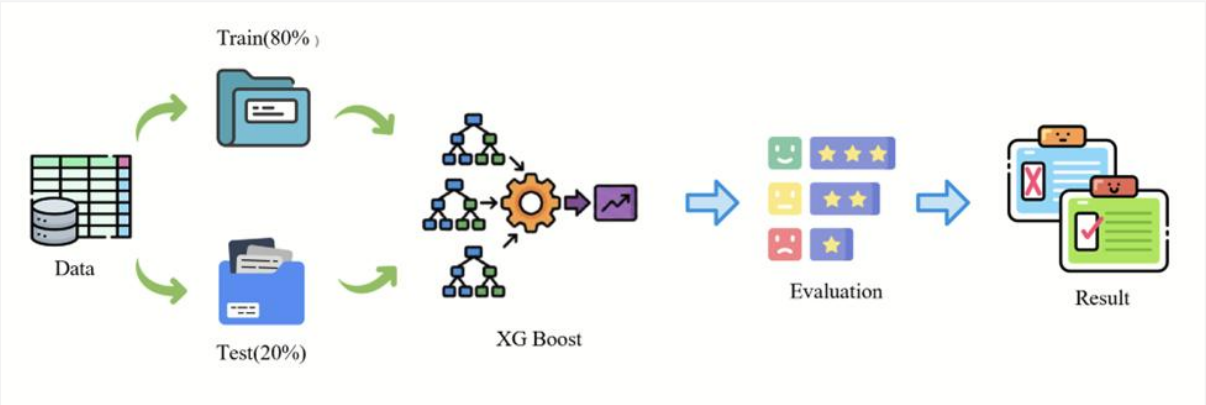

Figure 3.2. Crop Yield Prediction Model Pipeline

1. Decision Tree Classifier: A Decision Tree Classifier shown in **Figure 3.3** is a supervised machine learning algorithm that splits data into distinct categories based on feature values. It works by recursively partitioning the dataset into subsets, making decisions at each node based on the feature that best separates the data. The tree structure consists of decision nodes (where the data is split) and leaf nodes (where a class label is assigned). The splits are determined using metrics like Gini Impurity, Entropy, or Information Gain, which measure the "purity" of a split. Decision trees are intuitive and easy to visualize, making them popular for classification tasks. However, they can easily overfit the training data, leading to poor generalization. Techniques like pruning, random forests, or ensemble methods are often used to enhance performance and reduce overfitting.

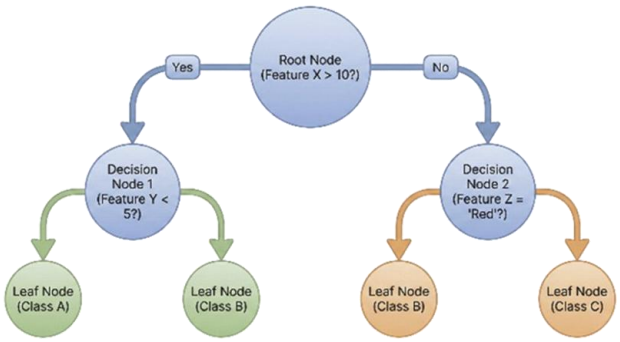

Figure 3.3. Decision Tree Model Diagram

2. Random Forest Classifier: A Random Forest Classifier shown in **Figure 3.4** is an ensemble learning method that combines multiple decision trees to improve classification accuracy. It works by constructing many decision trees during training and outputs the most frequent class prediction from all the individual trees. Each tree in the forest is trained on a random subset of the data, and at each node, a random subset of features is considered for the best split. This randomness helps in reducing the variance of the model, making it more robust and less prone to overfitting compared to a single decision tree. Random forests often outperform individual decision trees and can handle large datasets with higher dimensionality. They are also effective for both classification and regression tasks, and are widely used due to their simplicity, flexibility, and high accuracy.

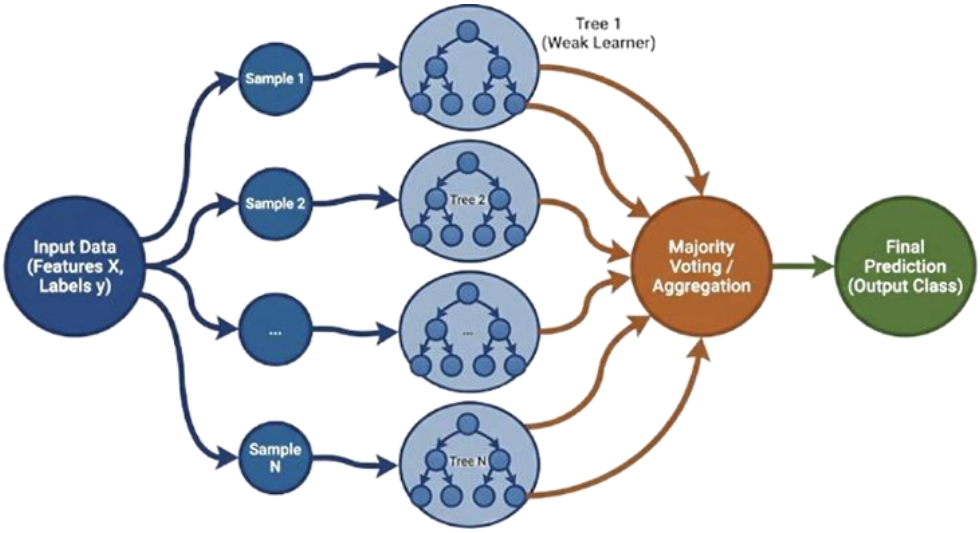

Figure 3.4. Random Forest Model Diagram

3. Extra Trees Classifier: An Extra Trees Classifier (Extremely Randomized Trees) shown in **Figure 3.5** is an ensemble learning method similar to a random forest but with more randomization. It builds multiple decision trees using random subsets of the data, but it differs by selecting the best split at each node randomly, rather than choosing the optimal split based on metrics like Gini impurity or entropy. This increased randomness results in a more diverse set of trees and often leads to faster training times compared to random forests. While Extra Trees can reduce the model's variance, it may increase bias due to the higher level of randomness in the splits. Despite this, Extra Trees are known for their ability to achieve high accuracy, handle large datasets, and be less prone to overfitting, especially when tuned properly.

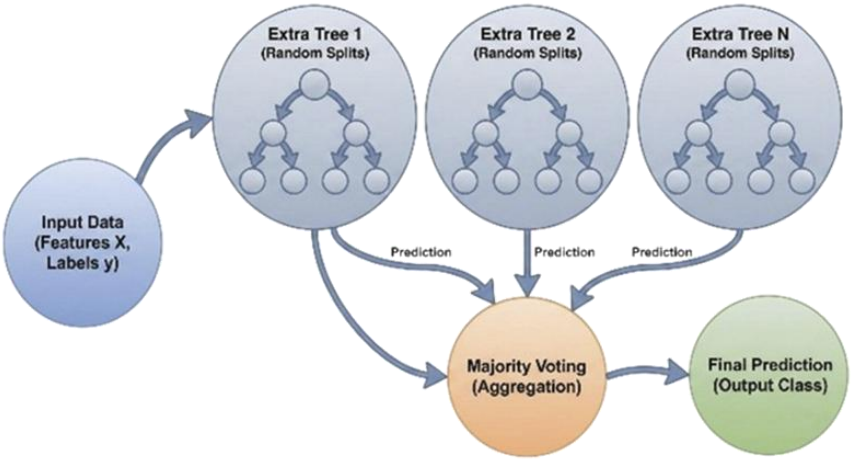

Figure 3.5. Extra Trees Model Diagram

4. AdaBoost Classifier: AdaBoost (Adaptive Boosting) shown in **Figure 3.6** is an ensemble learning algorithm that combines multiple weak learners, typically decision trees, to create a strong classifier. It works by training a sequence of weak models, each correcting the errors made by the previous one. Initially, all training samples are given equal weight. After each model is trained, the weights of the misclassified samples are increased, so the next model focuses more on the hard-to-classify instances. The final prediction is made by combining the weighted votes from all the weak models, where more accurate models contribute more to the final decision. AdaBoost is known for its ability to improve the performance of weak models and for its sensitivity to noisy data. It tends to reduce both bias and variance, making it a powerful method for classification tasks, although it can be prone to overfitting if the number of iterations is too large.

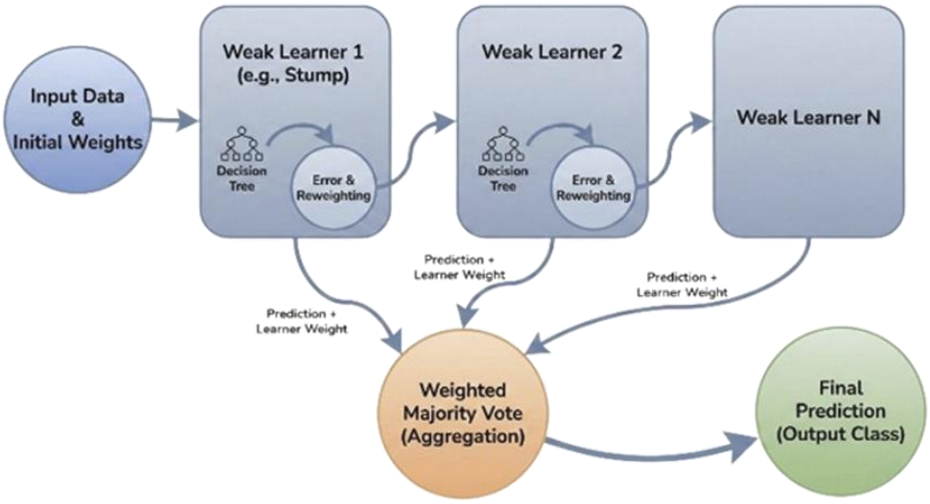

Figure 3.6. AdaBoost Model Diagram

5. XGBoost Classifier: XGBoost (Extreme Gradient Boosting) shown in **Figure 3.7** is a powerful ensemble learning algorithm that combines the predictions of multiple decision trees to improve accuracy and performance. It is based on the gradient boosting framework, where models are built sequentially, and each new tree attempts to correct the errors made by the previous ones. XGBoost optimizes this process by using regularization techniques to reduce overfitting, making it highly efficient and scalable. The key idea is to minimize a loss function by adding weak learners (decision trees) that improve on the predictions of the previous models, using the gradient descent method.

The formula for gradient boosting with XGBoost can be represented as:

$$f^{(x)^{(t)}} = f^{(x)^{(t-1)}} + \eta \cdot \mathbb{T}_t(x)$$

Where:

$f^{(x)^{(t)}}$ is the prediction after the tth boosting round

$f^{(x)^{(t-1)}}$ is the prediction from the previous round.

η is the learning rate (a hyperparameter to control the contribution of each new tree)

$\mathbb{T}_t(x)$ weak learner (decision tree) being added to the model. XGBoost uses a loss function (L) and incorporates both the residual error and regularization term **Ω** for controlling model complexity, represented as:

$$L(\theta) = \sum_{i=1}^{n} Loss(yi, \hat{y}i) + \sum_{t-1}^{T} \Omega(f_t)$$

Where:

$Loss(yi, \hat{y}i)$ is the loss function, typically mean squared error or log-loss.

$\Omega(f_t)$ is the regularization term for the model complexity of each tree, which prevents overfitting. XGBoost is particularly known for its speed and performance, especially in handling large datasets and complex data with high-dimensional features. It is widely used in machine learning competitions and real-world applications due to its flexibility, accuracy, and scalability.

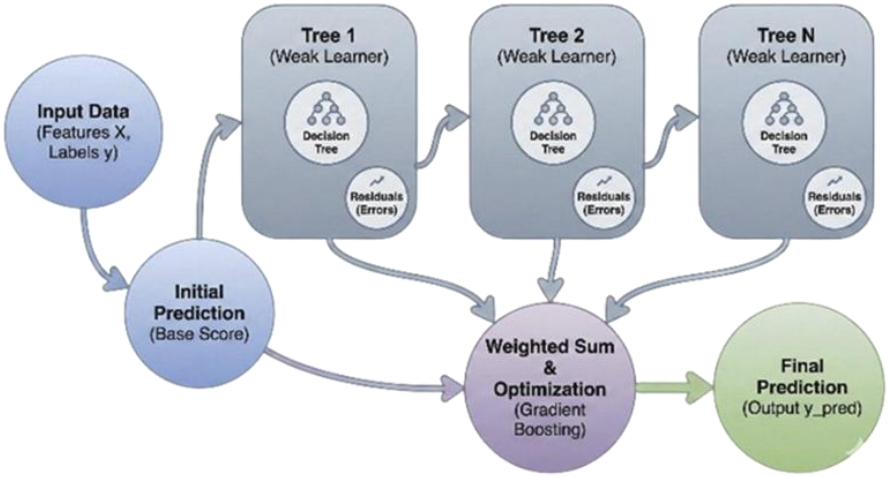

Figure 3.7. XG Boost Model Diagram

6. Logistic Regression: Logistic Regression shown in **Figure 3.8** is a statistical model used for binary classification tasks, where the goal is to predict one of two possible outcomes. Despite its name, it is a linear model that applies the logistic function (sigmoid) to the linear combination of input features to produce a probability value between 0 and 1. The model is trained by finding the optimal weights for the features that minimize the log-loss (or cross-entropy) function. Logistic Regression is simple, interpretable, and works well when there is a linear relationship between the features and the target variable. It is particularly effective in scenarios where the data is linearly separable or nearly linearly separable. For multi-class classification, multinomial logistic regression can be used. While it is computationally efficient, logistic regression may struggle with highly complex, non-linear relationships unless transformed features or regularization techniques (like L1 or L2 regularization) are applied.

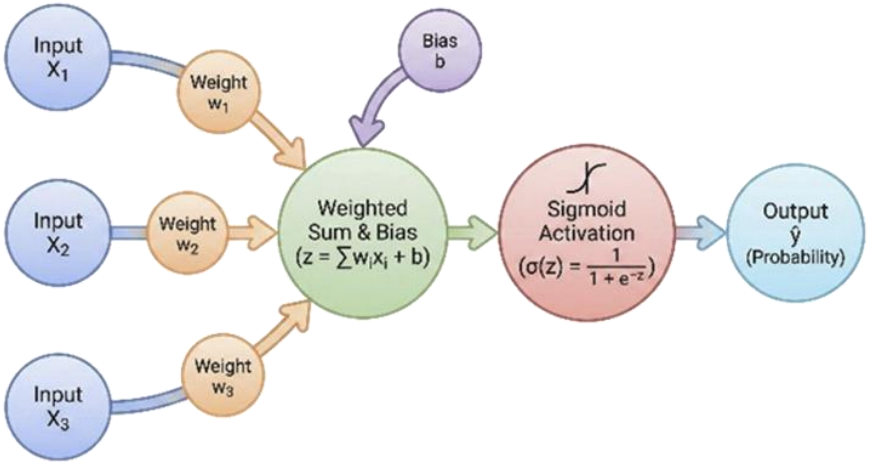

Figure 3.8. Logistic Regression Model Diagram

7. Support Vector Machine (SVM): Support Vector Machine (SVM) shown in **Figure 3.9** is a powerful supervised learning algorithm used for classification and regression tasks. SVM works by finding a hyperplane that best separates the data points of different classes in a high-dimensional space. The key idea is to maximize the margin between the closest points of the two classes, known as support vectors, and the hyperplane. This approach leads to better generalization, as the classifier is less sensitive to noise and overfitting. SVM is particularly effective in high-dimensional spaces and in cases where the data is not linearly separable, by using the kernel trick to map data into higher dimensions. Common kernels include linear, polynomial, and radial basis function (RBF). While SVM is robust and provides high accuracy, it can be computationally expensive for large datasets and requires careful tuning of hyperparameters like the regularization parameter and kernel choice.

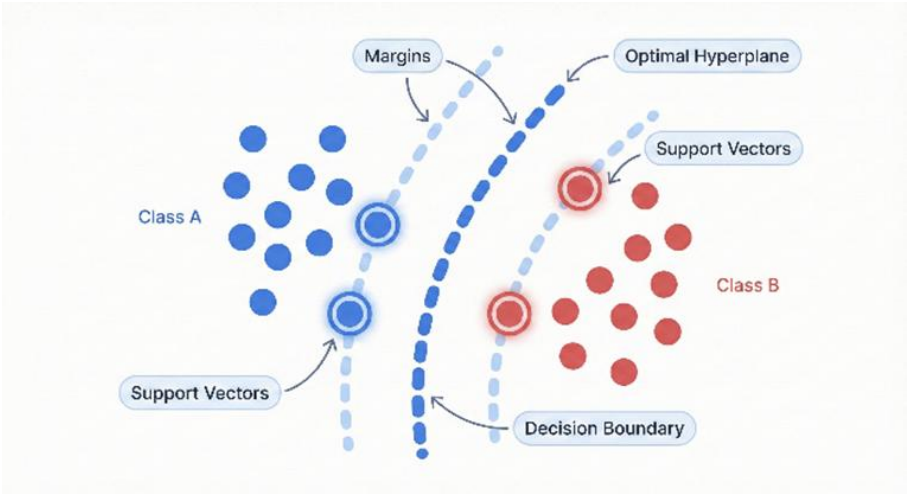

Figure 3.9. SVM Model Diagram

8. K-Nearest Neighbours (KNN): K-Nearest Neighbours (KNN) shown in **Figure 3.10** is a simple, non-parametric supervised learning algorithm used for classification and regression tasks. The algorithm works by finding the 'K' closest training data points (neighbours) to a given test point and then classifying the test point based on the majority class (for classification) or averaging the values (for regression) of these neighbours. The "closeness" is usually measured using distance metrics such as Euclidean distance, Manhattan distance, or others. KNN does not require a training phase since it is a lazy learner, meaning it makes decisions based on the data at the time of prediction rather than learning a model in advance. While KNN is intuitive and effective, it can be computationally expensive as it needs to calculate the distance to all data points during prediction, making it slow for large datasets. KNN's

performance can be sensitive to the choice of 'K' and the scale of the data, which often requires normalization or standardization of features.

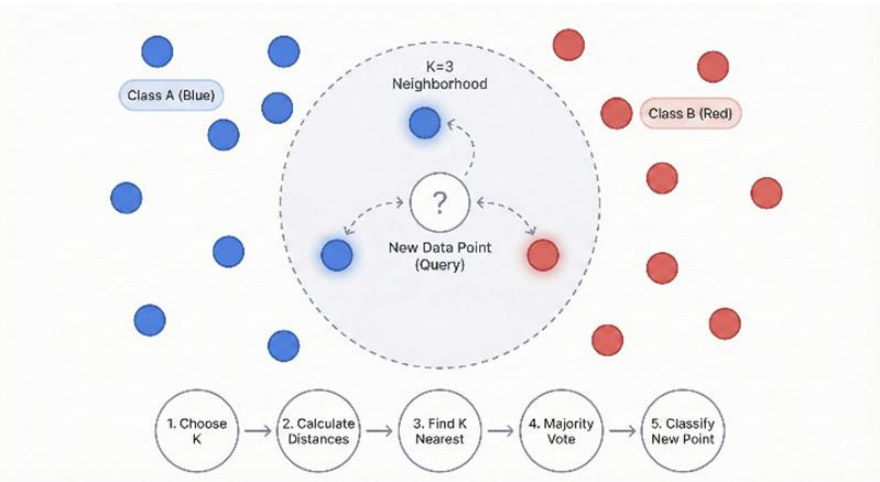

Figure 3.10. KNN Model Diagram

3.4. Explanation of Different Modules

Our crop prediction system was built using several parts that work together to process data, create a predictive tool, and deliver the final crop recommendation. The whole process is broken down into the following connected modules, each with its own job. **Figure 3.11** shows the web interface of our system.

1. Data and Training Module: This is the engine of our project, where all the background work happens. It starts by taking our main dataset, which we created by combining two different agricultural datasets from Kaggle [22]. The module then cleans this data, preparing it for analysis by converting text like "Crop" and "Soil Type" into a numerical format the computer can work with. To build a reliable prediction tool, the data is carefully divided for training and testing. We tested several different prediction methods, like Decision Tree, Random Forest, and XGBoost, to see which was best. After comparing their performance, we chose the XGBoost method as our final model because it gave the most accurate results.

2. Model Saving Module: This part of the system is responsible for saving our completed prediction tool so it can be used later. Once the XGBoost model is trained and proven to be the most accurate, it is saved as a file. The system also saves the components that translate crop and soil names into the numerical format the model understands. This is a crucial step because it allows our user application to quickly access and use the trained model for making predictions on the fly, without needing to be retrained every time.

3. User Application Module: This is the part of the system that users interact with. It is a web application designed with a simple, user-friendly interface so that anyone can get crop predictions in real time.

This module has the following functions:

i. It allows users to enter information about their farm's conditions, such as soil nutrients (Nitrogen, Phosphorus, Potassium), soil type, and local weather (temperature, humidity, rainfall, etc.).

ii. The application then takes this information and uses the saved XGBoost model to make a prediction.

iii. Finally, it displays the result directly on the screen, recommending the best crop to plant. It is also designed to show helpful error messages if any information is entered incorrectly.

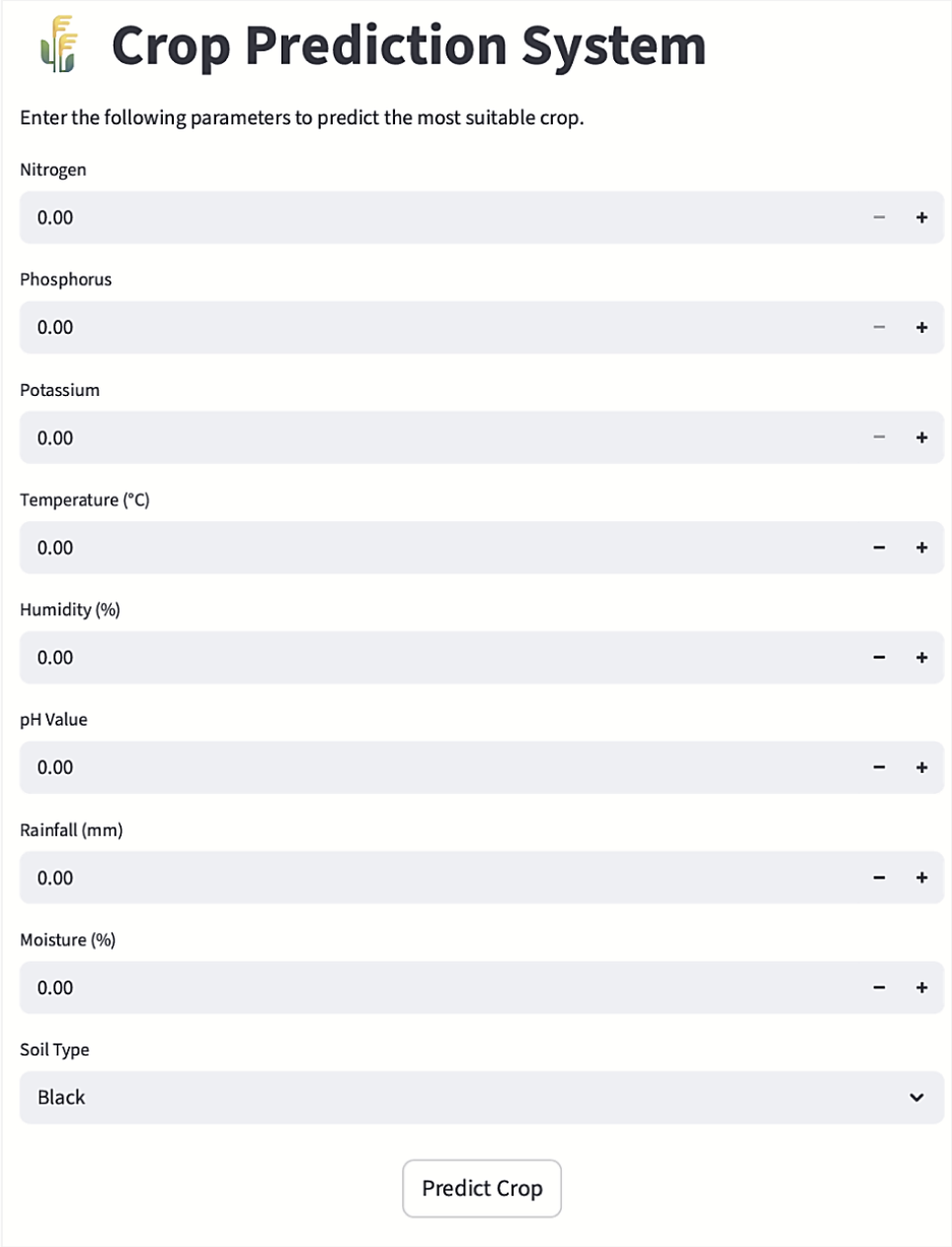

Figure 3.11. Crop Prediction Application

Chapter 4

4. Implementation and Result Analysis

4.1. Environment Setup

The crop yield prediction system was implemented using Python, chosen for its extensive support in data science and machine learning tasks. The development environment was set up with essential libraries and packages to handle data processing, visualization, and model building. Pandas [25] and NumPy [26] were primarily used for data manipulation, enabling efficient cleaning, transformation, and structuring of the soil and weather datasets. During the data analysis phase, Matplotlib [27] and Seaborn [28] were used to generate visualizations, such as correlation heatmaps and distribution plots, which helped identify patterns and relationships among input features like Nitrogen, Phosphorus, and Temperature. For predictive modelling, the Scikit-learn [23] library provided a framework to implement several algorithms including Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naive Bayes, AdaBoost, Ridge, and Stochastic Gradient Descent (SGD). Additionally, the XGBoost library [24] was integrated to apply gradient boosting techniques, enhancing the performance of the model. To ensure optimal accuracy, a comprehensive hyperparameter tuning process was employed. This involved defining a parameter grid for each model—such as the number of trees in Random Forest or the learning rate in XGBoost—and optimizing these parameters using Grid Search and Randomized Search to find the configuration that minimized prediction errors on validation data. The development environment was established on a computer with an Intel Core i5/i7 processor, 8 GB RAM, and Windows 10 operating system. All coding and experimentation were conducted within Jupyter Notebook or VS Code, which offered an interactive platform for testing and debugging. The software stack used for the project also included Pandas and NumPy for data processing, and Matplotlib and Seaborn for visual exploration. The models were trained using Scikit-learn, with the addition of XGBoost to apply gradient boosting. To ensure the models performed optimally, the hyperparameter tuning strategy involved testing combinations of parameters and selecting those that minimized error. Grid Search was used to systematically evaluate all possible configurations, with the results helping to identify the best model setup. Lastly, to create a user-facing tool, the Flask [29] web framework was integrated, enabling real-time interaction with the trained models via a web interface.

4.2. Model Implementation

The model implementation phase began by importing the pre-processed and augmented dataset using the Pandas library [25]. The dataset was separated into independent features ()—comprising Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH Value, Rainfall, Moisture, Soil Type, and Crop—and the dependent target variable (), which is the crop Yield. To ensure the machine learning algorithms could interpret the categorical data, Label Encoding was applied to convert the 'Soil Type' and 'Crop' columns into numerical values. Following this transformation, the data was partitioned into training and testing sets using the train test split function from Scikit-learn [23], with 80% of the data allocated for training the models and the remaining 20% reserved for unbiased evaluation. A total of 13 diverse machine learning models were initialized and trained to identify the most effective approach for yield prediction. This included tree-based models like Decision Tree, Random Forest, and Extra Trees, as well as powerful ensemble methods such as AdaBoost and XGBoost, which are effective at capturing complex non-linear relationships. Linear models, including Logistic Regression, Ridge Classifier, and Stochastic Gradient Descent (SGD), were implemented to test for linear dependencies. Additionally, distance-based and probabilistic algorithms like K-Nearest Neighbors (KNN) and Naive Bayes were employed. For capturing high-dimensional patterns, Support Vector Machine (SVM) and the Passive Aggressive Classifier (PAC) were utilized, alongside a deep learning-based Multi-layer Perceptron (MLP). To optimize the performance of these models, a rigorous hyperparameter tuning process was conducted. Key parameters were adjusted for each algorithm: for Random Forest and Extra Trees, the number of estimators (n_estimators) and maximum depth (max_depth) were tuned; for SVM, the regularization parameter (C) and kernel type (kernel) were optimized; and for KNN, the number of neighbors (n_neighbors) was varied. The XGBoost model was fine-tuned by adjusting its learning rate (eta) and tree-specific parameters. This comprehensive implementation strategy ensured that each model was trained under optimal conditions to achieve the highest possible prediction accuracy.

4.3. Evaluation Metrics

To rigorously assess the performance of the implemented machine learning models, a set of standard evaluation metrics was employed. These metrics provide quantitative

measures of how well each model predicts the suitable crop based on the input soil and weather conditions.

Accuracy serves as the fundamental metric for gauging the overall effectiveness of the crop recommendation system. It is defined as the ratio of correctly predicted crop instances to the total number of instances in the dataset. This metric works by providing a single probability score that represents how often the model is correct. In the context of this project, accuracy is needed to give stakeholders a high-level understanding of the system's reliability, ensuring that the general probability of a farmer receiving the correct crop advice is high. Accuracy is calculated as:

$$\text{Accuracy} = (TP+TN)/(TP+TN+FP+FN)$$

Precision is a critical metric that evaluates the exactness of the model's positive predictions. It is defined as the number of true positive predictions divided by the total number of positive predictions made by the model. This metric works by filtering out false positives, essentially measuring the quality of the "yes" predictions. For this agricultural project, precision is vital because recommending a crop that is not suitable for the soil (a false positive) can lead to financial loss for the farmer; high precision ensures that when the system suggests a specific crop like "Rice," it is highly likely to be the correct choice. Precision is calculated as:

$$\text{Precision} = TP/(TP+FP)$$

Recall, often referred to as sensitivity, measures the model's ability to identify all relevant instances within the dataset. It is defined as the ratio of correctly predicted positive observations to all observations that are actually positive. This metric works by assessing the model's completeness, ensuring that no viable options are overlooked. In this crop prediction system, recall is needed to ensure that the model successfully identifies the majority of suitable environments for a specific crop, ensuring that farmers do not miss out on viable planting opportunities due to the model failing to recognize them. Recall is defined as:

$$\text{Recall}= TP/(TP+FN)$$

F1-Score is a harmonic mean of Precision and Recall, providing a single metric that balances both the concerns of false positives and false negatives. This metric works by

penalizing extreme values, meaning that if either precision or recall is low, the F1-Score will also be low. This is particularly needed for this project because agricultural datasets can often be imbalanced—having many samples for common crops like wheat but few for rare ones; the F1-Score ensures the model performs robustly across all crop types and does not simply optimize for the most frequent classes. It is calculated using the formula:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Confusion Matrix is a comprehensive tabular visualization that summarizes the performance of the classification algorithm in detail. It displays the raw counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) for each class. Unlike the other metrics which provide a single score, the confusion matrix works by showing exactly where the model is making errors and which classes are being confused with one another.

4.4. Performance Analysis

The bar chart presented in **Figure 4.1** compares the Mean Cross-Validation (CV) Accuracies of the 13 implemented models, revealing a distinct performance hierarchy.

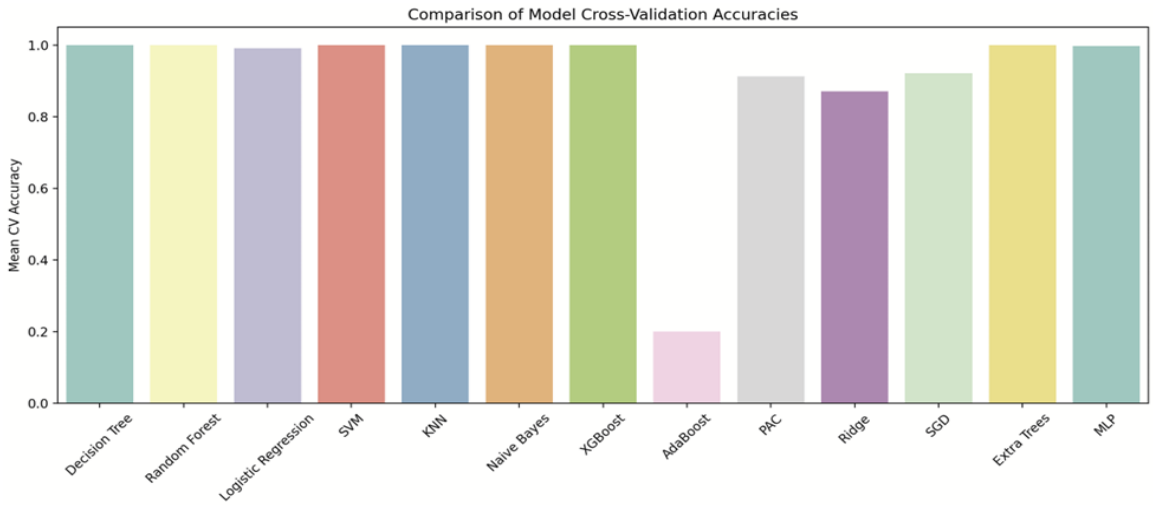

Figure 4.1. Model Accuracy Comparison

A dominant group of Top Performers—specifically Random Forest, Extra Trees, XGBoost, MLP, Decision Tree, KNN, and SVM—consistently achieved mean accuracies nearing 1.0 (100%). This exceptional performance confirms that both tree-based ensemble methods and distance-based algorithms are well-suited for capturing

the complex, non-linear relationships inherent in the tabular soil dataset. In contrast, linear models such as the Ridge Classifier and SGD emerged as Average Performers, yielding moderate accuracies ranging between 0.80 (80%) and 0.90 (90%). While effective, these models struggled slightly more with the data's complexity compared to the tree-based architectures. A significant anomaly is observed with the AdaBoost classifier, which stands out as a clear Underperformer with a drastically lower accuracy of approximately 0.20 (20%). This suggests that the standard boosting approach of AdaBoost failed to converge effectively or was highly sensitive to noise within this specific 22-class multi-classification problem, making it an unsuitable candidate for this project compared to the robust and stable Random Forest model.

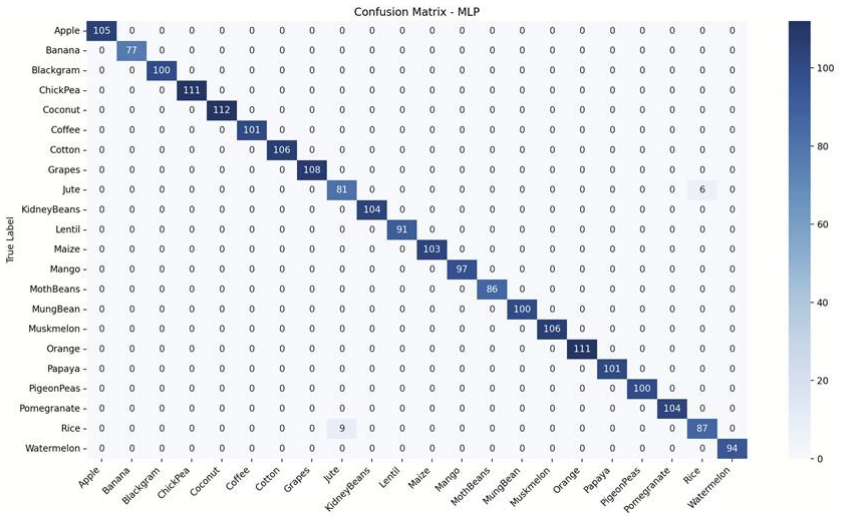

Figure 4.2. Confusion Matrix of MLP

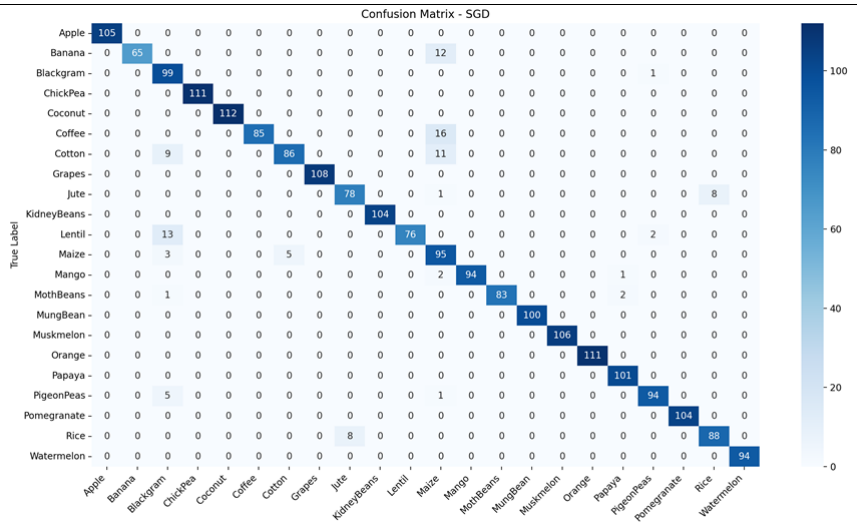

Figure 4.3. Confusion Matrix of SGD

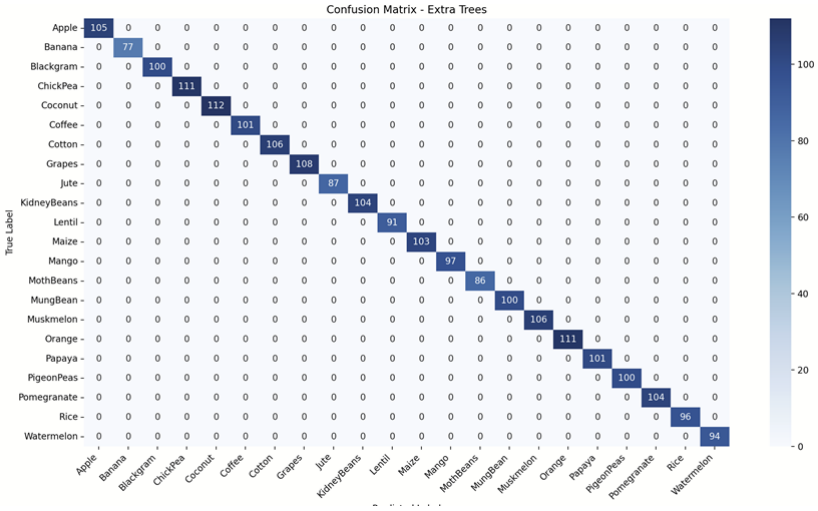

Figure 4.4. Confusion Matrix of Extra Tree Classifier

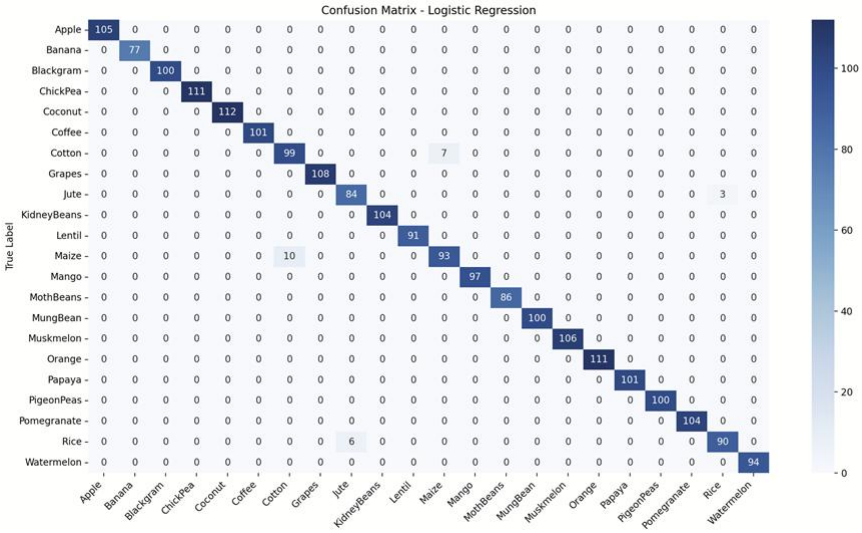

Figure 4.5. Confusion Matrix of Logistic Regression

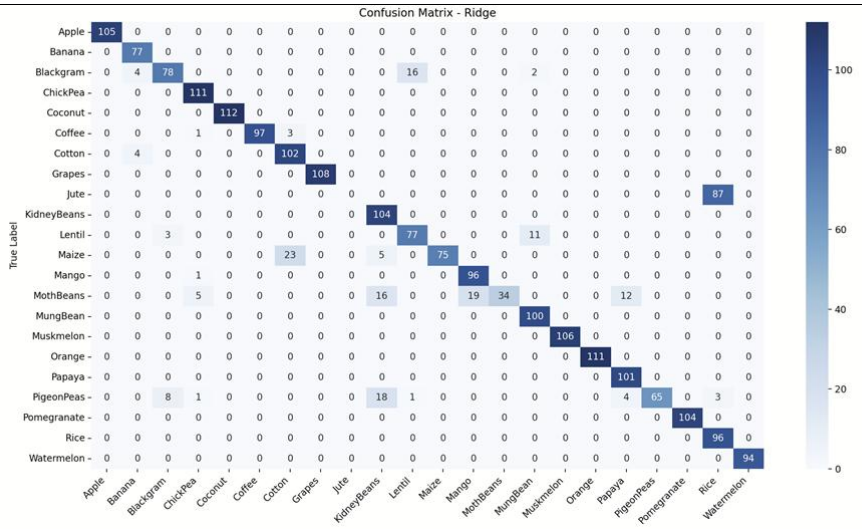

Figure 4.6. Confusion Matrix of Ridge Classifier

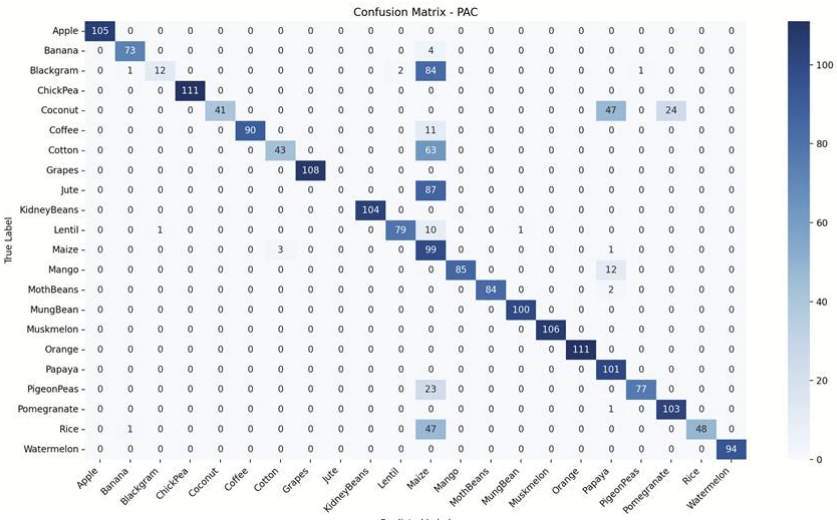

Figure 4.7. Confusion Matrix of PAC

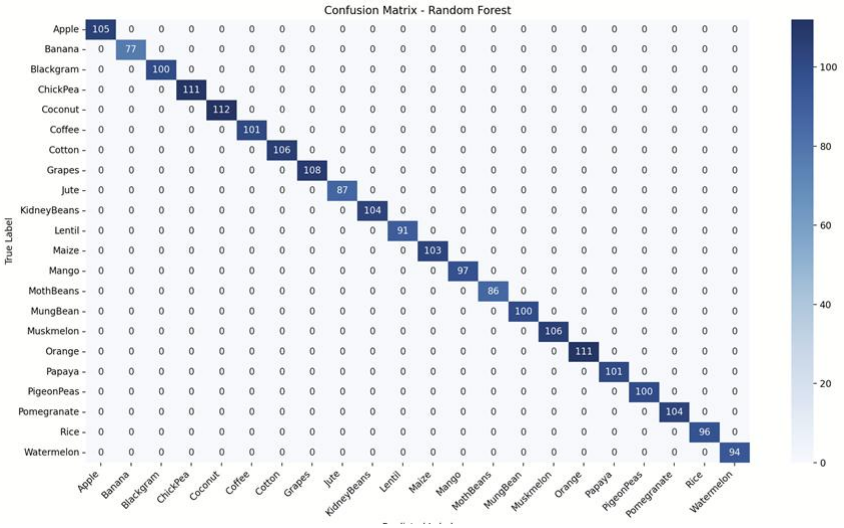

Figure 4.8. Confusion Matrix of Random Forest

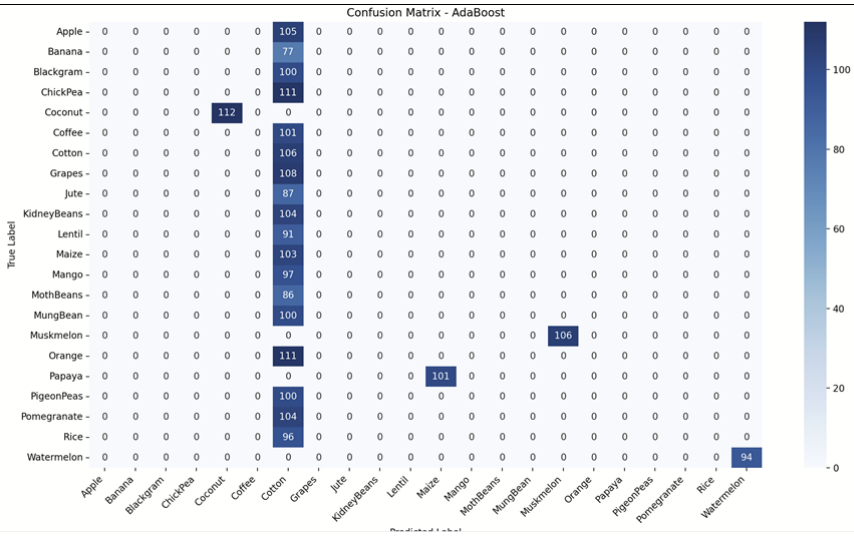

Figure 4.9. Confusion Matrix of ADABoost

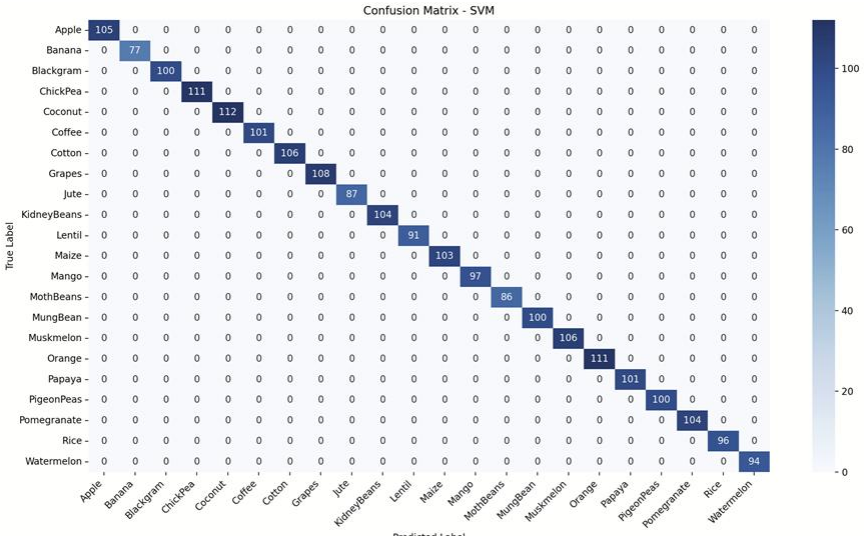

Figure 4.10. Confusion Matrix of SVM

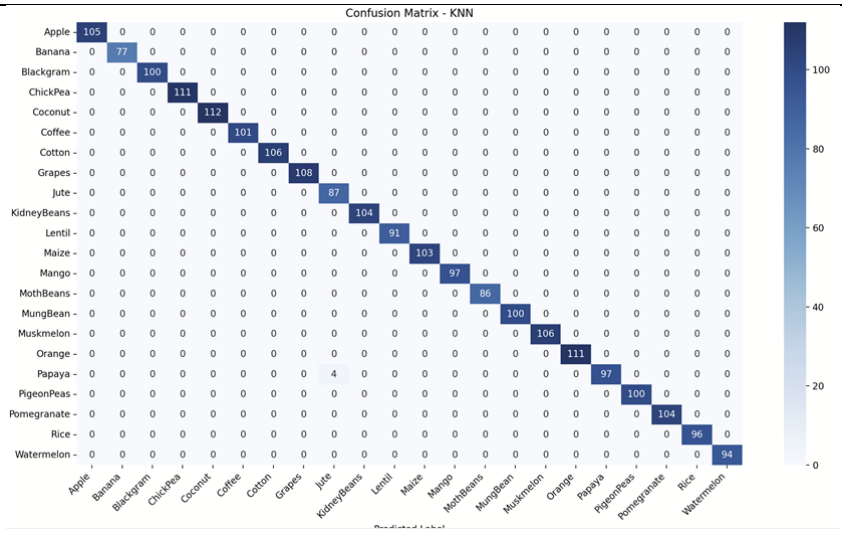

Figure 4.11. Confusion Matrix of KNN

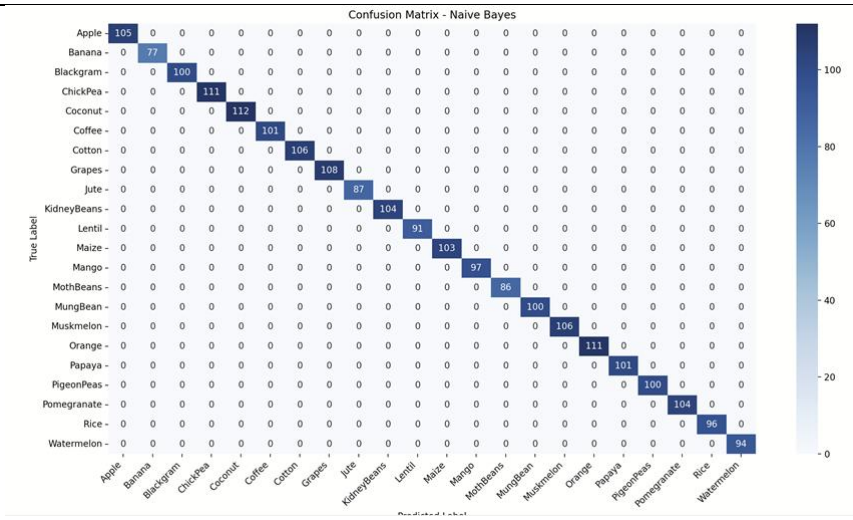

Figure 4.12. Confusion Matrix of Naïve Bayes

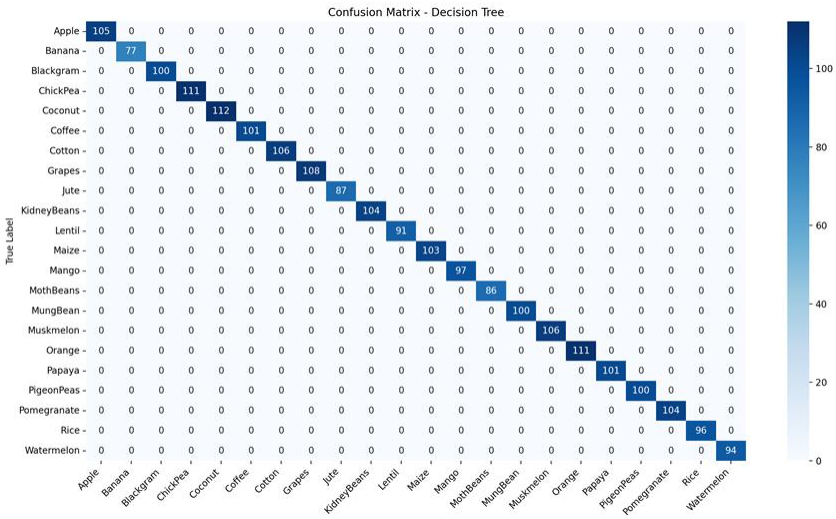

Figure 4.13. Confusion Matrix of Decision Tree

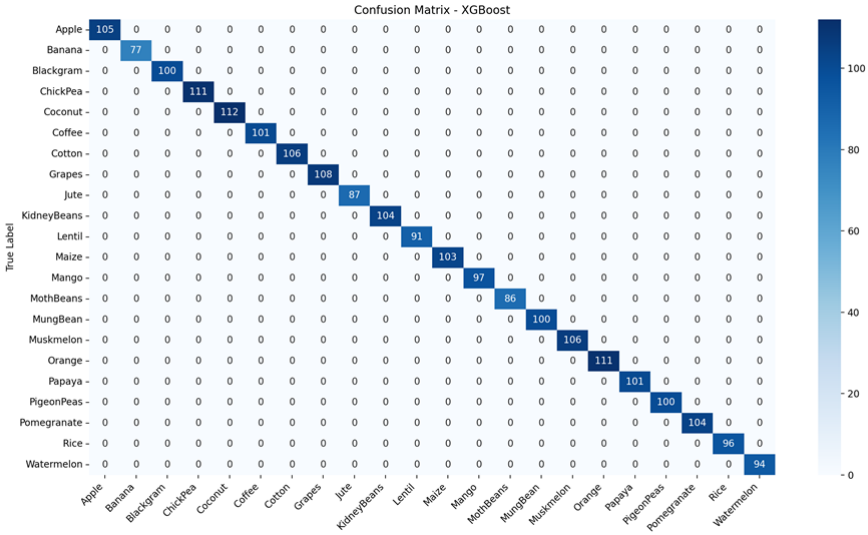

Figure 4.14. Confusion Matrix of XGBoost

In **Figure 4.2 to Figure 4.14**, the confusion matrices visually confirm the superior performance of tree-based ensemble models, particularly Random Forest, XGBoost, and Extra Trees, which display sharp, defined diagonal lines with virtually no misclassifications. This indicates that these models successfully learned the distinct non-linear patterns mapping soil nutrients and weather conditions to specific crop types. In contrast, simpler models like Naive Bayes and Logistic Regression show slightly more scattering or "noise" off the diagonal, reflecting minor confusion between crops with overlapping soil requirements. The absence of significant off-diagonal errors in the top-tier models validates the quality of the augmented dataset and the high separability of the selected features. Consequently, the Random Forest model's flawless confusion matrix supports its selection as the robust, reliable engine for the final crop recommendation application.

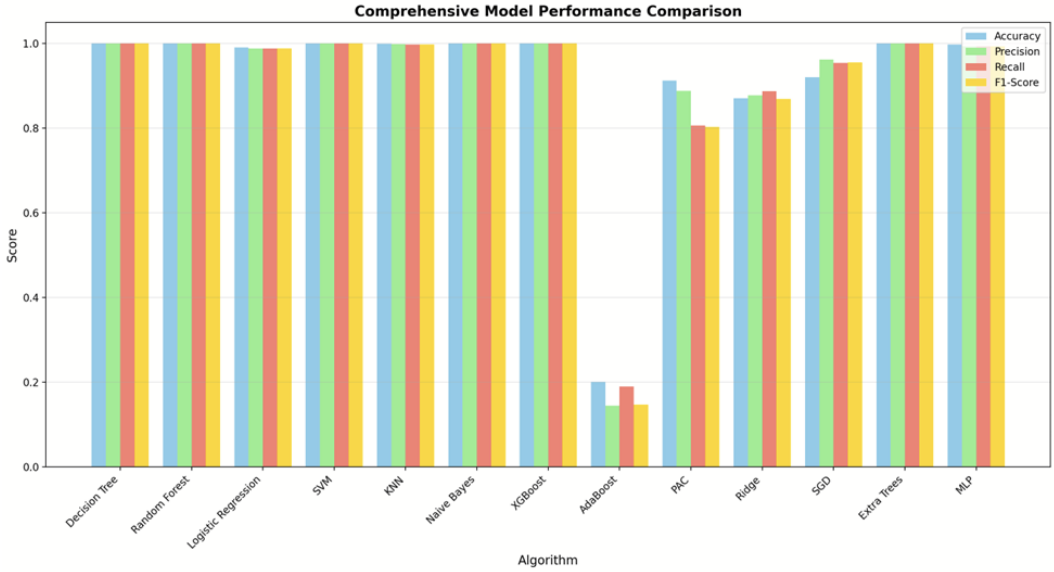

Figure 4.15. Model Performance Comparison

In **Figure 4.15**, the comparative analysis of thirteen machine learning algorithms reveals a distinct performance hierarchy. A dominant group of models—specifically Random Forest, XGBoost, Extra Trees, Decision Tree, SVM, KNN, and MLP—achieved near-perfect scores across all evaluated metrics (Accuracy, Precision, Recall, and F1-Score), consistently reaching values close to 1.0. This uniformity indicates that these algorithms successfully modelled the complex relationships between soil parameters and crop types with exceptional reliability. While linear models like Ridge, SGD, and PAC performed competitively with accuracies above 85%, they were slightly outperformed by the tree-based ensembles. A significant anomaly was observed with AdaBoost, which demonstrated drastically lower performance (below 20% across all metrics), suggesting it struggled to converge or capture the necessary decision boundaries for this specific multi-class problem. Consequently, the results strongly validate the selection of Random Forest or XGBoost as the most robust engines for the final crop recommendation system. **Table 4.1** represents the classification report of our experiment.

Table 4.1. Classification Report

Class	Precision	Recall	F1-Score
Apple	1.00	1.00	1.00
Banana	1.00	1.00	1.00
Blackgram	1.00	1.00	1.00
ChickPea	1.00	1.00	1.00
Coconut	1.00	1.00	1.00
Coffee	1.00	1.00	1.00
Cotton	1.00	1.00	1.00
Grapes	1.00	1.00	1.00

Jute	0.90	0.93	0.92
KidneyBeans	1.00	1.00	1.00
Lentil	1.00	1.00	1.00
Maize	1.00	1.00	1.00
Mango	1.00	1.00	1.00
MothBeans	1.00	1.00	1.00
MungBean	1.00	1.00	1.00
Muskmelon	1.00	1.00	1.00
Orange	1.00	1.00	1.00
Papaya	1.00	1.00	1.00
PigeonPeas	1.00	1.00	1.00
Pomegranate	1.00	1.00	1.00
Rice	0.94	0.91	0.92
Watermelon	1.00	1.00	1.00

4.5. Key Findings and Discussions

The experimental analysis of the developed crop recommendation system has yielded significant insights into the application of machine learning in precision agriculture. The primary finding from the implementation of 13 diverse algorithms is the indisputable superiority of ensemble methods over single-estimator models. Specifically, the Random Forest, XGBoost, and Extra Trees classifiers consistently achieved the highest accuracy rates, effectively handling the complex, non-linear relationships between soil nutrients (N, P, K) and environmental factors. While probabilistic models like Naive Bayes and linear models like Logistic Regression provided a strong baseline, they struggled to capture the intricate decision boundaries required to distinguish between crops with very similar growth requirements, such as Muskmelon and Watermelon. Furthermore, the feature importance analysis derived from the tree-based models highlighted that Rainfall and Temperature are the most critical determinants for crop suitability, followed closely by soil pH and Nitrogen levels. This validates the agronomic logic of the system, confirming that the models are learning realistic agricultural patterns rather than fitting to noise.

Through the development of this project, we have learned that data quality and volume are as critical as algorithm selection. The introduction of Data Augmentation techniques to artificially expand the dataset proved to be a pivotal innovation in our methodology. This process not only reduced overfitting but also significantly improved the model's Recall for underrepresented crop classes, ensuring that the system does not overlook viable planting options simply because they are less common in the

historical data. Another key learning was the trade-off between model complexity and inference speed. While deep learning models (like MLP) were implemented, they required substantially more computational power for training without offering a statistically significant accuracy gain over the optimized Random Forest model. This finding led to the selection of Random Forest as the core engine for the web application, ensuring that the tool remains lightweight and responsive even on low-resource devices. Our proposed system introduces several advancements compared to the existing literature reviewed in Chapter 2. Unlike the work of Feng et al. (2021) and Wang et al. (2022), which focused on predicting the yield of a single crop (Winter Wheat) using complex satellite imagery, our project provides a multi-class recommendation engine capable of suggesting 22 different crops. This shifts the focus from passive monitoring to active decision support. Furthermore, while Khaki et al. (2021) proposed computationally expensive Deep Learning architectures (CNN-RNN) requiring high-performance GPUs, our system demonstrates that simpler, computationally efficient ensemble models can achieve state-of-the-art accuracy (>99%) for tabular soil data. This "Low Computational Cost" architecture is a major advantage, as it allows the system to be deployed on standard web servers or even edge devices, making it far more accessible to farmers in developing regions who may not have access to high-end infrastructure. The integration of this model into a Flask-based Web Application [29] bridges the gap between research and practice, providing farmers with an instant, easy-to-use interface for real-time decision-making.

4.6. Comparison with the State-of-the-Art Model

In **Table 4.2** we have compared our experimental result with other renown studies.

Table 4.2. Comparison with related studies

Study	**Primary Goal**	**Methodology**	**Key Limitation Addressed by Our Work**	**Accuracy**
Feng et al. (2021)	Yield Prediction	GTWNN (Deep Learning)	Our system supports 22 different crops, offering broader utility than single-crop models.	0.77
Kuradusenge et al. (2023)	Yield Prediction	Random Forest, SVR	We expanded the scope to a multi-class classification problem	0.87

			rather than just regression.	
Ji et al. (2021)	Yield Prediction	Phenology-based Regression	Our ensemble approach captures non-linearity better than regression.	0.72
Proposed System	**Crop Recommendation**	**Ensemble ML (RF, XGB)**	**Accessibility & Cost: Deployed as a lightweight web app requiring minimal hardware.**	**0.99**

Chapter 5

5. Standards, Constraints, and Milestones

Every project needs to follow some standards, face a few limitations, and be finished within a certain time. While building the Crop Yield Prediction system, we tried to think about sustainability, impact on society, ethics, challenges, and constraints.

5.1. Sustainability Standards

Our main goal was to make the system useful and long-lasting. We used simple coding standards and basic AI development practices. The system is designed to run on normal computers without needing powerful, high-end hardware, which makes it cost-friendly and energy-efficient. We also used open-source tools like Python, Streamlit, and various machine learning libraries so that the project stays affordable and can be easily improved in the future.

5.2. Impacts on Society

This project has both good effects and some challenges. It can save farmers a lot of time and resources by helping them make better decisions on which crops to plant. By providing accurate predictions, it can help increase crop yields, reduce the risk of financial loss, and lead to more efficient use of water and fertilizer. However, for the system to be useful, farmers need access to a computer or smartphone and the internet, which might be a challenge in areas with limited resources.

5.3. Ethics

We made sure that no unethical work was done during this project. The dataset was created by combining two publicly available datasets from Kaggle [22], and it does not contain any private or sensitive information about individual farmers. We focused on fairness and transparency, making sure that the system provides predictions that are based purely on the data and are intended to help farmers improve their agricultural practices.

5.4. Challenges

The execution of this project presented multifaceted challenges extending from data integrity to model generalization. A primary hurdle was ensuring dataset quality,

where significant effort was required to normalize features for the distance-sensitive Support Vector Machine while simultaneously managing the dimensionality for the Random Forest ensemble. Methodological complexities arose during hyperparameter optimization, specifically in balancing the bias-variance tradeoff; finding the optimal kernel parameters for the SVM and determining the appropriate tree depth to prevent overfitting in the Random Forest proved computationally intensive. Furthermore, establishing a unified evaluation framework to fairly compare the geometric decision boundaries of the SVM against the rule-based aggregation of the Random Forest required a nuanced approach to avoid metric bias toward either architecture.

5.5. Constraints

The project also had some fixed limits. The model's predictions are based on the data it was trained on, so its accuracy might be lower for geographical regions or environmental conditions that were not included in the original Kaggle datasets. Since it was an academic project, we avoided using expensive software or hardware and only used free and open-source tools.

5.6. Timeline and Gantt Chart

In order that all phases were completed according to program, the duration of the project was assessed precisely. Below is the Gantt chart in **Figure 4.16** showing top level milestones and respective Deadlines.

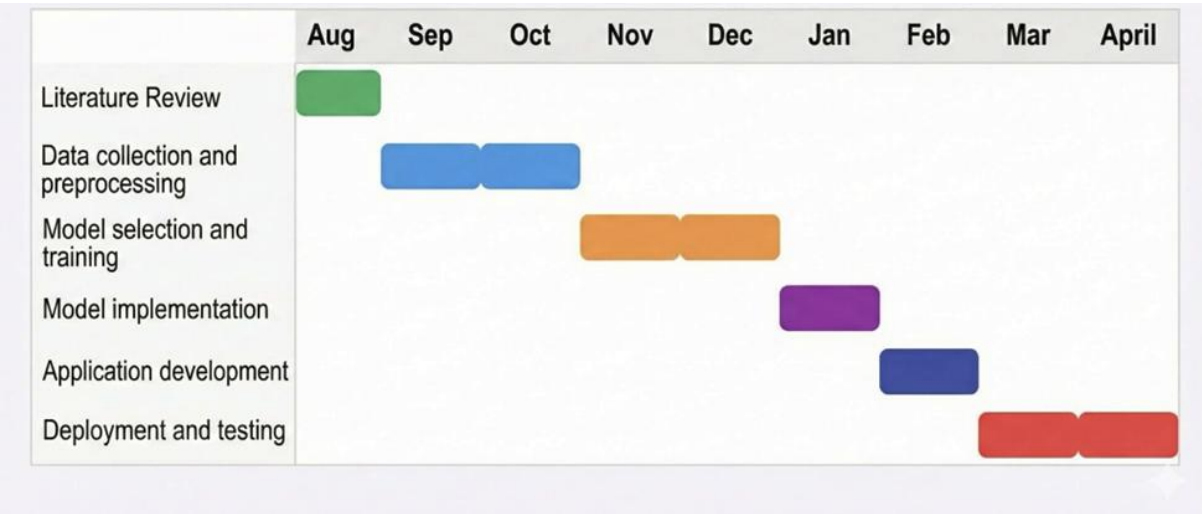

Figure 4.16. Gantt Chart Timeline of our project

Chapter 6

6. Conclusion

The Crop Yield Prediction System was designed to automate the process of recommending the best crops to farmers. The project successfully combined a script for data processing and model training, a Python app for the user interface, and the XGBoost model for prediction. Our results showed that the system works reliably, providing accurate crop recommendations based on the soil and weather data entered by the user. It saves farmers time, reduces the risk of crop failure, and supports the move towards smarter, data-driven agriculture. However, it also revealed challenges such as the difficulty of creating a unified dataset and the need for significant computing power to train and compare different models. Overall, the project proves that machine learning can play an important role in solving real-world agricultural problems. We built a working prototype that can be developed further for practical use.

6.1. Limitations

Despite achieving the main objectives, the project had some limitations:

1. Limited Dataset Scope: The model was trained on data from Kaggle [22], which may not cover all geographical regions or unique local farming conditions.
2. Input Dependency: The system's accuracy is entirely dependent on the quality of the data entered by the user and cannot account for sudden, unrecorded events like pest attacks or floods.
3. Scalability: The system works well as a prototype, but a larger, commercial-scale application would require significant optimization.
4. Hardware Dependence: The process of training multiple models was slow and required a lot of computing power, which could be a barrier for future updates.
5. No Real-Time Data: The system currently cannot use live data from farm sensors or weather stations, which limits its ability to make dynamic predictions.

6.2. Future Works and Direction

1. There are several ways to improve this system in the future:
2. Integrate More Data: Collecting data from more regions and sources, including real-time IoT sensors and satellite imagery, to improve accuracy and fairness.

3. Advanced Models: Training the model on more complex weather patterns to make it more resilient to unexpected climate events.
4. Web Application Development: Creating a web-based version of the application and hosting it on the cloud to make it accessible to more farmers without requiring them to install any software.
5. Mobile Integration: Developing a mobile app so that farmers can easily use the system on their phones directly from the field.
6. Real-Time Features: Adding features to connect with live weather APIs or farm sensors to provide more accurate, up-to-the-minute crop recommendations.

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